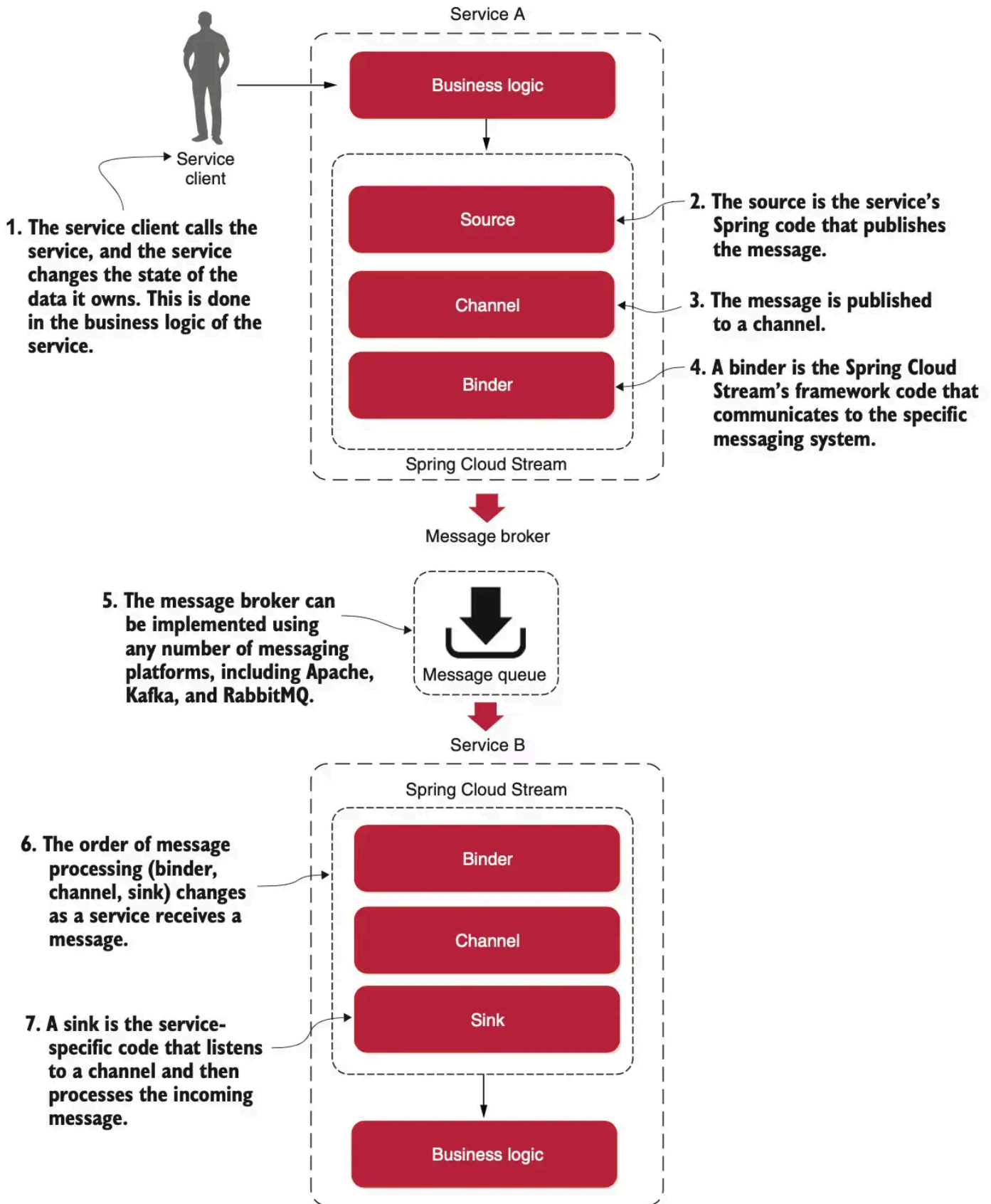


Asynchronous communications (Spring Cloud Stream)

Introduction

[Spring Cloud Stream](#) allows to abstract away the implementation details of the messaging platform (e.g., Apache Kafka, RabbitMQ). Thus, **implementation-specific details are kept out of the application code**. The publication and consumption of messages in applications is done through platform-neutral Spring interfaces.

Let's begin our discussion by looking at the Spring Cloud Stream architecture through the lens of two services communicating via messaging. One service is the **message publisher**, and one service is the **message consumer**.



- **Source:** Converts a POJO into a serialized message (default JSON) and publishes it to a channel.
- **Channel:** An abstraction over a queue; decouples the application from the actual queue name, allowing configuration changes without code changes.
- **Binder:** Connects to a specific messaging platform, letting the application send and receive messages without using platform-specific APIs.
- **Sink:** Listens on a channel, deserializes messages back into POJOs, and passes them to the service's business logic.

Project dependencies

Add `spring-cloud-stream`

Add at least one binder (e.g., `spring-cloud-starter-stream-rabbit` or `spring-cloud-starter-stream-kafka`).

The `<dependencyManagement>` section is typically used in Spring-based projects to manage Spring Cloud dependencies consistently. By using the `spring-cloud-dependencies` [BOM](#), you can ensure that the correct dependencies are used and avoid version conflicts.

```
<properties>
  <spring-cloud.version>2025.1.0</spring-cloud.version>
</properties>

<dependencies>
  <dependency>
    <groupId>org.springframework.cloud</groupId>
    <artifactId>spring-cloud-stream</artifactId>
  </dependency>
  <dependency>
    <groupId>org.springframework.cloud</groupId>
    <artifactId>spring-cloud-starter-stream-rabbit</artifactId>
  </dependency>
</dependencies>

<dependencyManagement>
  <dependencies>
    <dependency>
      <groupId>org.springframework.cloud</groupId>
      <artifactId>spring-cloud-dependencies</artifactId>
      <version>${spring-cloud.version}</version>
      <type>pom</type>
      <scope>import</scope>
    </dependency>
  </dependencies>
</dependencyManagement>
```

Resources

- <https://www.baeldung.com/spring-maven-bom>